$C_{\phi d}^{(22)}\text{i1\_i2\_} \to$

$$
\begin{aligned}
&\frac{1}{16\pi^2}\Big(-\frac{2}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_d^p]\,\delta_{i1i2}+\frac{5}{108}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_d^p]\,\delta_{i1i2}-\frac{8}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_d^p]\,\delta_{i1i2}-\\
&\frac{2}{27}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_e^p]\,\delta_{i1i2}+\frac{5}{36}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_e^p]\,\delta_{i1i2}-\frac{8}{135}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_e^p]\,\delta_{i1i2}-\\
&\frac{1}{27}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_l^p]\,\delta_{i1i2}+\frac{5}{72}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_l^p]\,\delta_{i1i2}-\frac{4}{135}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_l^p]\,\delta_{i1i2}-\\
&\frac{1}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_q^p]\,\delta_{i1i2}+\frac{5}{216}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_q^p]\,\delta_{i1i2}-\frac{4}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_q^p]\,\delta_{i1i2}-\\
&\frac{8}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_u^p]\,\delta_{i1i2}+\frac{5}{27}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_u^p]\,\delta_{i1i2}-\frac{32}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_u^p]\,\delta_{i1i2}-\\
&\frac{1}{27}g_1^4\,\mathrm{LF}_{3,0}[\tilde\mu]\,\delta_{i1i2}-\frac{1}{18}g_1^4\,\mathrm{LF}_{4,-1}[\tilde\mu]\,\delta_{i1i2}+\frac{8}{135}g_1^4\,\mathrm{LF}_{5,-2}[\tilde\mu]\,\delta_{i1i2}-\\
&\frac{1}{324}g_1^4\,\mathrm{LF}_{2,1,0}[m_1,m_d^{i1}]\,\delta_{i1i2}-\frac{1}{324}g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,m_d^{i1}]\,\delta_{i1i2}+\\
&\frac{1}{162}g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,m_d^{i1}]\,\delta_{i1i2}-\frac{1}{324}g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,m_d^{i1}]\,\delta_{i1i2}-\\
&\frac{1}{324}g_1^4\,\mathrm{LF}_{2,1,0}[m_1,m_d^{i2}]\,\delta_{i1i2}-\frac{1}{324}g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,m_d^{i2}]\,\delta_{i1i2}+\\
&\frac{1}{162}g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,m_d^{i2}]\,\delta_{i1i2}-\frac{1}{324}g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,m_d^{i2}]\,\delta_{i1i2}-\\
&\frac{1}{36}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,0}[m_1,m_q^p]+\frac{1}{18}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{3,1,-1}[m_1,m_q^p]-\\
&\frac{1}{36}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{4,1,-2}[m_1,m_q^p]-\frac{3}{4}g_2^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,0}[m_2,m_q^p]+\\
&\frac{3}{2}g_2^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{3,1,-1}[m_2,m_q^p]-\frac{3}{4}g_2^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{4,1,-2}[m_2,m_q^p]-\\
&\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}[m_3,m_d^{i1}]\,\delta_{i1i2}-\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{2,2,-1}[m_3,m_d^{i1}]\,\delta_{i1i2}+\\
&\frac{2}{27}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}[m_3,m_d^{i1}]\,\delta_{i1i2}-\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{4,1,-2}[m_3,m_d^{i1}]\,\delta_{i1i2}-\\
&\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}[m_3,m_d^{i2}]\,\delta_{i1i2}-\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{2,2,-1}[m_3,m_d^{i2}]\,\delta_{i1i2}+\\
&\frac{2}{27}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}[m_3,m_d^{i2}]\,\delta_{i1i2}-\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{4,1,-2}[m_3,m_d^{i2}]\,\delta_{i1i2}-\\
&\frac{4}{3}g_3^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,0}[m_3,m_q^p]+\frac{8}{3}g_3^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{3,1,-1}[m_3,m_q^p]-\\
&\frac{4}{3}g_3^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{4,1,-2}[m_3,m_q^p]-\frac{1}{9}g_1^2\,\overline{a_d}^{pr}a_d^{pr}\,\mathrm{LF}_{2,2,0}[m_d^r,m_q^p]\,\delta_{i1i2}+\\
&\frac{1}{18}g_1^2\,\overline{a_d}^{pr}a_d^{pr}\,\mathrm{LF}_{3,2,-1}[m_d^r,m_q^p]\,\delta_{i1i2}+\frac{1}{162}g_1^4\,\mathrm{LF}_{2,1,0}[m_d^{i1},m_1]\,\delta_{i1i2}-\\
&\frac{1}{324}g_1^4\,\mathrm{LF}_{3,1,-1}[m_d^{i1},m_1]\,\delta_{i1i2}+\frac{2}{27}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}[m_d^{i1},m_3]\,\delta_{i1i2}-\\
&\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}[m_d^{i1},m_3]\,\delta_{i1i2}+\frac{1}{162}g_1^4\,\mathrm{LF}_{2,1,0}[m_d^{i2},m_1]\,\delta_{i1i2}-\\
&\frac{1}{324}g_1^4\,\mathrm{LF}_{3,1,-1}[m_d^{i2},m_1]\,\delta_{i1i2}+\frac{2}{27}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}[m_d^{i2},m_3]\,\delta_{i1i2}-\\
&\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}[m_d^{i2},m_3]\,\delta_{i1i2}-\frac{1}{9}g_1^2\,\overline{a_e}^{pr}a_e^{pr}\,\mathrm{LF}_{2,2,0}[m_e^r,m_l^p]\,\delta_{i1i2}+\\
&\frac{1}{18}g_1^2\,\overline{a_e}^{pr}a_e^{pr}\,\mathrm{LF}_{3,2,-1}[m_e^r,m_l^p]\,\delta_{i1i2}+\frac{1}{9}g_1^2\,\overline{a_e}^{pr}a_e^{pr}\,\mathrm{LF}_{3,1,0}[m_l^p,m_e^r]\,\delta_{i1i2}+\\
&\frac{1}{9}g_1^2\,\overline{a_e}^{pr}a_e^{pr}\,\mathrm{LF}_{3,2,-1}[m_l^p,m_e^r]\,\delta_{i1i2}-\frac{1}{4}g_1^2\,\overline{a_e}^{pr}a_e^{pr}\,\mathrm{LF}_{4,1,-1}[m_l^p,m_e^r]\,\delta_{i1i2}+\\
&\frac{1}{9}g_1^2\,\overline{a_e}^{pr}a_e^{pr}\,\mathrm{LF}_{5,1,-2}[m_l^p,m_e^r]\,\delta_{i1i2}+\frac{1}{9}g_1^2\,\overline{a_d}^{pr}a_d^{pr}\,\mathrm{LF}_{3,1,0}[m_q^p,m_d^r]\,\delta_{i1i2}+\\
&\frac{1}{9}g_1^2\,\overline{a_d}^{pr}a_d^{pr}\,\mathrm{LF}_{3,2,-1}[m_q^p,m_d^r]\,\delta_{i1i2}-\frac{5}{12}g_1^2\,\overline{a_d}^{pr}a_d^{pr}\,\mathrm{LF}_{4,1,-1}[m_q^p,m_d^r]\,\delta_{i1i2}+\\
&\frac{1}{3}g_1^2\,\overline{a_d}^{pr}a_d^{pr}\,\mathrm{LF}_{5,1,-2}[m_q^p,m_d^r]\,\delta_{i1i2}+\frac{1}{18}g_1^2\tilde\mu^2\,\overline{y_u}^{pr}y_u^{pr}\,\mathrm{LF}_{2,2,0}[m_q^p,m_u^r]\,\delta_{i1i2}-\\
&\frac{1}{36}g_1^2\tilde\mu^2\,\overline{y_u}^{pr}y_u^{pr}\,\mathrm{LF}_{3,2,-1}[m_q^p,m_u^r]\,\delta_{i1i2}-\frac{1}{18}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,0}[m_q^p,\tilde\mu]+\\
&\frac{1}{36}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,2,-1}[m_q^p,\tilde\mu]+\frac{1}{36}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{3,1,-1}[m_q^p,\tilde\mu]-\\
&\frac{1}{18}g_1^2\tilde\mu^2\,\overline{y_u}^{pr}y_u^{pr}\,\mathrm{LF}_{3,1,0}[m_u^r,m_q^p]\,\delta_{i1i2}-\frac{1}{18}g_1^2\tilde\mu^2\,\overline{y_u}^{pr}y_u^{pr}\,\mathrm{LF}_{3,2,-1}[m_u^r,m_q^p]\,\delta_{i1i2}-\\
&\frac{1}{6}g_1^2\tilde\mu^2\,\overline{y_u}^{pr}y_u^{pr}\,\mathrm{LF}_{4,1,-1}[m_u^r,m_q^p]\,\delta_{i1i2}+\frac{1}{3}g_1^2\tilde\mu^2\,\overline{y_u}^{pr}y_u^{pr}\,\mathrm{LF}_{5,1,-2}[m_u^r,m_q^p]\,\delta_{i1i2}+\\
&\frac{5}{36}g_1^4\,\mathrm{LF}_{4,1,-2}[\tilde\mu,m_1]\,\delta_{i1i2}-\frac{1}{9}g_1^4\,\mathrm{LF}_{5,1,-3}[\tilde\mu,m_1]\,\delta_{i1i2}+\frac{5}{12}g_1^2g_2^2\,\mathrm{LF}_{4,1,-2}[\tilde\mu,m_2]\,\delta_{i1i2}-\\
&\frac{1}{3}g_1^2g_2^2\,\mathrm{LF}_{5,1,-3}[\tilde\mu,m_2]\,\delta_{i1i2}-\frac{1}{2}\overline{y_d}^{ps}\,\overline{y_d}^{ri1}y_d^{pi2}y_d^{rs}\,\mathrm{LF}_{2,1,0}[\tilde\mu,m_d^s]+\\
&\overline{y_d}^{ps}\,\overline{y_d}^{ri1}y_d^{pi2}y_d^{rs}\,\mathrm{LF}_{3,1,-1}[\tilde\mu,m_d^s]-\frac{1}{2}\overline{y_d}^{ps}\,\overline{y_d}^{ri1}y_d^{pi2}y_d^{rs}\,\mathrm{LF}_{4,1,-2}[\tilde\mu,m_d^s]+\\
&\frac{1}{36}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,0}[\tilde\mu,m_q^p]+\frac{7}{36}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{3,1,-1}[\tilde\mu,m_q^p]-\\
&\frac{1}{18}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{4,1,-2}[\tilde\mu,m_q^p]-\frac{1}{2}\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\,\mathrm{LF}_{2,1,0}[\tilde\mu,m_u^s]+\\
&\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\,\mathrm{LF}_{3,1,-1}[\tilde\mu,m_u^s]-\frac{1}{2}\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\,\mathrm{LF}_{4,1,-2}[\tilde\mu,m_u^s]-\\
&\frac{1}{36}m_1g_1^2y_d^{pi2}\,\overline{a_d}^{pi1}\,\mathrm{LF}_{2,1,1,0}[m_1,m_d^{i1},m_q^p]+\frac{1}{72}m_1g_1^2y_d^{pi2}\,\overline{a_d}^{pi1}\,\mathrm{LF}_{2,2,1,-1}[m_1,m_d^{i1},m_q^p]+\\
&\frac{1}{36}m_1g_1^2y_d^{pi2}\,\overline{a_d}^{pi1}\,\mathrm{LF}_{3,1,1,-1}[m_1,m_d^{i1},m_q^p]-\frac{1}{72}g_1^4\,\mathrm{LF}_{2,1,1,-1}[m_1,m_d^{i1},\tilde\mu]\,\delta_{i1i2}-\\
&\frac{1}{36}g_1^4m_1^2\,\mathrm{LF}_{2,1,1,0}[m_1,m_d^{i1},\tilde\mu]\,\delta_{i1i2}-\frac{1}{36}m_1g_1^2\,\overline{y_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{2,1,1,0}[m_1,m_d^{i2},m_q^p]+\\
&\frac{1}{72}m_1g_1^2\,\overline{y_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{2,2,1,-1}[m_1,m_d^{i2},m_q^p]+\frac{1}{36}m_1g_1^2\,\overline{y_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{3,1,1,-1}[m_1,m_d^{i2},m_q^p]-\\
&\frac{1}{72}g_1^4\,\mathrm{LF}_{2,1,1,-1}[m_1,m_d^{i2},\tilde\mu]\,\delta_{i1i2}-\frac{1}{36}g_1^4m_1^2\,\mathrm{LF}_{2,1,1,0}[m_1,m_d^{i2},\tilde\mu]\,\delta_{i1i2}-\\
&\frac{1}{72}m_1g_1^2y_d^{pi2}\,\overline{a_d}^{pi1}\,\mathrm{LF}_{2,2,1,-1}[m_1,m_q^p,m_d^{i1}]-\frac{1}{72}m_1g_1^2\,\overline{y_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{2,2,1,-1}[m_1,m_q^p,m_d^{i2}]+\\
&\frac{1}{4}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,-1}[m_1,m_q^p,\tilde\mu]-\frac{1}{6}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{3,1,1,-2}[m_1,m_q^p,\tilde\mu]-\\
&\frac{1}{72}g_1^2\left(6\,\overline{y_d}^{pi1}y_d^{pi2}+g_1^2\delta_{i1i2}\right)\mathrm{LF}_{2,2,1,-2}[m_1,\tilde\mu,m_d^{i1}]+\frac{1}{72}g_1^4m_1^2\\
&\quad\mathrm{LF}_{2,2,1,-1}[m_1,\tilde\mu,m_d^{i1}]\,\delta_{i1i2}-\frac{1}{72}g_1^2\left(6\,\overline{y_d}^{pi1}y_d^{pi2}+g_1^2\delta_{i1i2}\right)\mathrm{LF}_{2,2,1,-2}[m_1,\tilde\mu,m_d^{i2}]+\\
&\frac{1}{72}g_1^4m_1^2\,\mathrm{LF}_{2,2,1,-1}[m_1,\tilde\mu,m_d^{i2}]\,\delta_{i1i2}-\frac{5}{24}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,2,1,-2}[m_1,\tilde\mu,m_q^p]+\\
&\frac{9}{4}g_2^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,-1}[m_2,m_q^p,\tilde\mu]-\frac{3}{2}g_2^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{3,1,1,-2}[m_2,m_q^p,\tilde\mu]-\\
&\frac{9}{8}g_2^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,2,1,-2}[m_2,\tilde\mu,m_q^p]+\frac{2}{3}m_3g_3^2y_d^{pi2}\,\overline{a_d}^{pi1}\,\mathrm{LF}_{2,1,1,0}[m_3,m_d^{i1},m_q^p]-\\
&\frac{1}{3}m_3g_3^2y_d^{pi2}\,\overline{a_d}^{pi1}\,\mathrm{LF}_{2,2,1,-1}[m_3,m_d^{i1},m_q^p]-\frac{2}{3}m_3g_3^2y_d^{pi2}\,\overline{a_d}^{pi1}\,\mathrm{LF}_{3,1,1,-1}[m_3,m_d^{i1},m_q^p]+\\
&\frac{2}{3}m_3g_3^2\,\overline{y_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{2,1,1,0}[m_3,m_d^{i2},m_q^p]-\frac{1}{3}m_3g_3^2\,\overline{y_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{2,2,1,-1}[m_3,m_d^{i2},m_q^p]-\\
&\frac{2}{3}m_3g_3^2\,\overline{y_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{3,1,1,-1}[m_3,m_d^{i2},m_q^p]+\frac{1}{3}m_3g_3^2y_d^{pi2}\,\overline{a_d}^{pi1}\,\mathrm{LF}_{2,2,1,-1}[m_3,m_q^p,m_d^{i1}]+\\
&\frac{1}{3}m_3g_3^2\,\overline{y_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{2,2,1,-1}[m_3,m_q^p,m_d^{i2}]-\frac{1}{8}\tilde\mu^2\,\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\,\mathrm{LF}_{2,2,1,-1}[m_q^p,\tilde\mu,m_u^s]-\\
&\frac{1}{8}\tilde\mu^2\,\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\,\mathrm{LF}_{2,2,1,-1}[m_q^r,\tilde\mu,m_u^s]+\\
&\frac{1}{8}\tilde\mu^2\,\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\,\mathrm{LF}_{2,2,1,-1}[m_u^s,\tilde\mu,m_q^p]+\\
&\frac{1}{8}\tilde\mu^2\,\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\,\mathrm{LF}_{2,2,1,-1}[m_u^s,\tilde\mu,m_q^r]+\frac{1}{4}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,-1}[\tilde\mu,m_1,m_d^{i1}]-\\
&\frac{1}{6}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{3,1,1,-2}[\tilde\mu,m_1,m_d^{i1}]+\frac{1}{4}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,-1}[\tilde\mu,m_1,m_d^{i2}]-\\
&\frac{1}{6}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{3,1,1,-2}[\tilde\mu,m_1,m_d^{i2}]-\frac{1}{8}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,-1}[\tilde\mu,m_1,m_q^p]-\\
&\frac{3}{8}g_2^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,-1}[\tilde\mu,m_2,m_q^p]+\frac{1}{4}\tilde\mu^2\,\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\,\mathrm{LF}_{2,1,1,0}[\tilde\mu,m_q^p,m_u^s]-\\
&\frac{1}{4}\tilde\mu^2\,\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\,\mathrm{LF}_{3,1,1,-1}[\tilde\mu,m_q^p,m_u^s]+\frac{1}{4}\tilde\mu^2\,\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\\
&\quad\mathrm{LF}_{2,1,1,0}[\tilde\mu,m_q^r,m_u^s]-\frac{1}{4}\tilde\mu^2\,\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\,\mathrm{LF}_{3,1,1,-1}[\tilde\mu,m_q^r,m_u^s]+\frac{1}{12}m_1g_1^2y_d^{pi2}\\
&\quad\overline{a_d}^{pi1}\,\mathrm{LF}_{1,1,1,1,0}[m_1,m_d^{i1},m_q^p,\tilde\mu]+\frac{1}{24}m_1g_1^2y_d^{pi2}\,\overline{a_d}^{pi1}\,\mathrm{LF}_{2,1,1,1,-1}[m_1,m_d^{i1},m_q^p,\tilde\mu]+\\
&\frac{1}{12}m_1g_1^2\,\overline{y_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{1,1,1,1,0}[m_1,m_d^{i2},m_q^p,\tilde\mu]+\frac{1}{24}m_1g_1^2\,\overline{y_d}^{pi1}a_d^{pi2}\\
&\quad\mathrm{LF}_{2,1,1,1,-1}[m_1,m_d^{i2},m_q^p,\tilde\mu]-\frac{1}{4}\overline{y_d}^{si1}y_d^{pi2}\,\overline{a_d}^{pr}a_d^{sr}\,\mathrm{LF}_{2,1,1,1,-1}[m_d^r,m_q^p,m_q^s,\tilde\mu]+\\
&\frac{1}{24}m_1g_1^2y_d^{pi2}\,\overline{a_d}^{pi1}\,\mathrm{LF}_{2,1,1,1,-1}[m_d^{i1},m_1,m_q^p,\tilde\mu]+\frac{1}{24}m_1g_1^2\,\overline{y_d}^{pi1}a_d^{pi2}\\
&\quad\mathrm{LF}_{2,1,1,1,-1}[m_d^{i2},m_1,m_q^p,\tilde\mu]+\frac{1}{18}g_1^2\,\overline{a_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}[m_q^p,m_1,m_d^{i1},m_d^{i2}]-\\
&\frac{1}{24}m_1g_1^2y_d^{pi2}\,\overline{a_d}^{pi1}\,\mathrm{LF}_{2,1,1,1,-1}[m_q^p,m_1,m_d^{i1},\tilde\mu]-\frac{1}{24}m_1g_1^2\,\overline{y_d}^{pi1}a_d^{pi2}\\
&\quad\mathrm{LF}_{2,1,1,1,-1}[m_q^p,m_1,m_d^{i2},\tilde\mu]+\frac{2}{3}g_3^2\,\overline{a_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}[m_q^p,m_3,m_d^{i1},m_d^{i2}]+\\
&\frac{1}{4}\tilde\mu^2\,\overline{y_d}^{pi1}y_d^{ri2}\,\overline{y_u}^{rs}y_u^{ps}\,\mathrm{LF}_{2,1,1,1,-1}[m_u^s,m_q^p,m_q^r,\tilde\mu]-\frac{1}{24}m_1g_1^2y_d^{pi2}\,\overline{a_d}^{pi1}\\
&\quad\mathrm{LF}_{2,1,1,1,-1}[\tilde\mu,m_1,m_d^{i1},m_q^p]-\frac{1}{24}m_1g_1^2\,\overline{y_d}^{pi1}a_d^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}[\tilde\mu,m_1,m_d^{i2},m_q^p]\Big)
\end{aligned}
$$